Research Report: OCR-VLM Approach for Document Intelligence

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1. Executive Summary

OCR technology is evolving from traditional multi-stage pipelines toward **Vision-Language Model (VLM)-based document understanding**. Traditional OCR typically separates text detection, text recognition, and layout analysis into independent stages. This can create multiple failure points, especially when processing tables, mathematical equations, multi-column layouts, handwriting, forms, and complex visual elements.

The OCR-VLM approach treats a document page as a visual input and uses a vision-language model to understand both **what the content says and where it appears on the page**. Depending on the architecture, the model can produce Markdown, HTML, JSON, LaTeX, bounding boxes, semantic tags, or other structured representations.

This report reviews five open-source OCR-VLM systems in the following order:

1. **PaddleOCR-VL**
2. **Nanonets OCR 2**
3. **DeepSeek-OCR**
4. **OlmOCR-2**
5. **LightOn OCR**

These systems optimize different aspects of document processing:

- **PaddleOCR-VL:** compact multilingual VLM with controlled layout and content processing
- **Nanonets OCR 2:** semantic document tagging
- **DeepSeek-OCR:** visual-token compression
- **OlmOCR-2:** reinforcement learning based on document-level correctness
- **LightOn OCR:** compact, end-to-end architecture designed for fine-tuning

The research demonstrates that next-generation document intelligence systems should evaluate OCR not only on character recognition but also on **layout preservation, semantic understanding, structured extraction, multilingual support, visual grounding, document-level correctness, and computational efficiency**.

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2. Research Objective

The objective of this research is to understand modern **OCR-VLM architectures** and identify architectural principles that can be applied to a document intelligence system.

The research focuses on:

- Vision-Language Model architecture for OCR
- Document layout understanding
- Text recognition
- Table extraction
- Formula recognition
- Image and chart understanding
- Spatial grounding
- Semantic document representation
- Vision-token efficiency
- Multilingual document processing
- End-to-end versus hybrid VLM architectures
- Fine-tuning and domain adaptation
- Document-level evaluation

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3. Evolution from Traditional OCR to OCR-VLM

3.1 Traditional OCR Pipeline

A conventional OCR system commonly consists of several independent stages:

```
Document
   ↓
Image Preprocessing
   ↓
Text Detection
   ↓
Text Recognition
   ↓
Layout Analysis
   ↓
Table Detection
   ↓
Formula Recognition
   ↓
Post-processing
```

```
    ↓
Final Text
```

Each component specializes in a particular task.

This architecture can work well for relatively simple documents, but complex documents introduce several failure points.

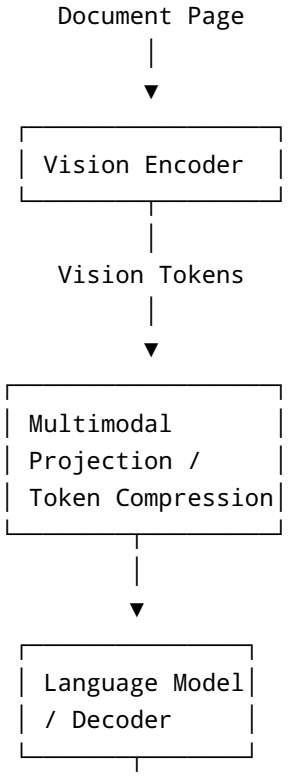
For example:

- A table can be detected incorrectly.
- Text from multiple columns can be read in the wrong order.
- Mathematical notation can lose spatial relationships.
- A signature can be interpreted as ordinary text.
- Images can be separated from the text describing them.
- Handwritten information can be difficult for conventional recognition systems.
- Document structure can be lost during post-processing.

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4. OCR-VLM Architecture

A simplified OCR-VLM architecture can be represented as:



```
                    |
                    ▼
            Structured Output
```

The important difference is that the model can reason about **visual content and language simultaneously**.

Instead of simply asking:

> "What characters are present?"

the system can reason about:

> "What is this element, where is it located, how is it related to surrounding elements, and how should it be represented in structured output?"

This capability is especially important for document intelligence applications.

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5. Major Architectural Challenge: Vision Tokens

One of the central problems in OCR-VLM systems is the number of visual tokens generated from a document page.

A document page can contain a large amount of information. High-resolution images preserve small fonts, mathematical symbols, table borders, and degraded text, but high resolution can produce a large number of vision tokens.

More tokens generally mean:

- Higher computational requirements
- Higher memory requirements
- Slower inference
- Greater attention-computation requirements

However, aggressive visual compression can remove important information.

Therefore, OCR-VLM architecture must balance:

```
Resolution
   ↕
Vision Tokens
   ↕
Accuracy
```

```
   ↕
Memory
   ↕
Inference Speed
```

This trade-off is particularly important for high-resolution documents, long PDFs, and large-scale processing.

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6. Model Architecture Research

6.1 PaddleOCR-VL

PaddleOCR-VL — Hugging Face

Architecture

PaddleOCR-VL uses a compact architecture combining:

- NaViT-style dynamic-resolution vision encoder
- ERNIE-4.5-0.3B language model

The model is designed for resource-efficient document parsing. It supports 109 languages and handles text, tables, formulas, and charts.

Hybrid Architecture

Unlike a completely end-to-end architecture, PaddleOCR-VL uses a two-stage strategy:

```
Document
   ↓
PP-DocLayoutV2
   ↓
Layout Detection
   ↓
Document Regions
   ↓
PaddleOCR-VL
   ↓
Region Recognition
```

The first stage identifies document elements and their locations.

The second stage processes the relevant regions using task-specific prompts such as:

```
OCR:
Table Recognition:
Formula Recognition:
Chart Recognition:
```

Key Innovation

The major innovation is **compact multilingual document intelligence**.

The system supports 109 languages, including scripts such as:

- Chinese
- English
- Japanese
- Korean
- Russian
- Arabic
- Hindi/Devanagari
- Thai

Research Insight

PaddleOCR-VL demonstrates an important alternative to pure end-to-end VLMs:

> Use a lightweight layout detector to control what the VLM sees.

This can reduce unnecessary processing while maintaining explicit control over difficult document elements.

6.2 Nanonets OCR 2

Nanonets OCR 2 — Hugging Face

Architecture

Nanonets OCR 2 is based on a fine-tuned Qwen2.5-VL architecture and is designed specifically for document-to-structured-output conversion.

The model is designed for a broad range of document categories, including:

- Research documents
- Financial reports
- Legal contracts
- Healthcare records

- Tax forms
- Receipts
- Invoices
- Tables
- Handwriting
- Signatures
- Watermarks
- Checkboxes
- Charts
- Embedded images

Key Innovation: Semantic Tagging

Nanonets OCR 2 goes beyond plain text extraction by introducing semantic tags.

Examples include:

```
<signature>
<watermark>
<page_number>
<img>
<checkbox>
```

This allows downstream systems to distinguish between different document elements rather than receiving one large block of text.

The architecture can also represent:

- Tables as HTML
- Equations as LaTeX
- Images through image tags and descriptions
- Flowcharts and organizational charts through structured representations
- Checkbox states using standardized symbols

Document Intelligence Value

This approach is particularly useful when OCR output will immediately enter an LLM or structured extraction pipeline.

For example:

```
Document
   ↓
OCR-VLM
   ↓
```

```
Semantic Document Representation
   ↓
LLM / Information Extraction
   ↓
Structured JSON
```

Research Insight

Nanonets OCR 2 demonstrates that OCR output should not necessarily be treated as plain text. Semantic document markup can provide a valuable intermediate representation for downstream AI systems.

6.3 DeepSeek-OCR

DeepSeek-OCR — Hugging Face

Architecture

DeepSeek-OCR uses a custom architecture focused heavily on **visual-token compression**.

The architecture consists of:

```
Document Image
      ↓
   SAM-base
      ↓
Token Compression
      ↓
  CLIP-large
      ↓
DeepSeek-3B MoE Decoder
```

The architecture includes:

- DeepEncoder
- A visual-token compression module
- A DeepSeek mixture-of-experts decoder

DeepEncoder

The DeepEncoder combines:

1. SAM-base
2. A token compressor

3. CLIP-large

The token compressor significantly reduces the number of vision tokens passed to the language model.

Resolution Modes

DeepSeek-OCR provides multiple resolution modes:

```
Tiny
Small
Base
Large
Gundam
Gundam-Master
```

These modes allow the system to trade image detail against computational cost.

Key Innovation

The major research contribution is **visual-token efficiency**.

Rather than simply increasing model size, DeepSeek-OCR attempts to represent document information using a smaller visual-token representation.

Research Insight

DeepSeek-OCR highlights an important research direction:

OCR-VLM performance may depend as much on efficient visual representation as on language-model size.

6.4 OlmOCR-2

OlmOCR-2 — Hugging Face

Architecture

OlmOCR-2 is fine-tuned from Qwen2.5-VL-7B-Instruct. It also uses GRPO reinforcement learning to improve difficult OCR cases such as equations and tables.

Training Architecture

The system uses two major training stages:

```
Qwen2.5-VL-7B
       ↓
Supervised Fine-Tuning
       ↓
olmOCR-mix
       ↓
GRPO Reinforcement Learning
       ↓
olmOCR-2
```

The reinforcement-learning stage uses generated document examples and programmatic tests to evaluate whether generated outputs satisfy document-level requirements.

Key Innovation

Traditional OCR evaluation often relies on character-level or token-level similarity.

OlmOCR-2 instead emphasizes **document-level correctness**.

For example, mathematical output can be evaluated based on whether the rendered equation is correct rather than simply whether the generated LaTeX string resembles a reference string.

Research Insight

This introduces an important concept:

OCR-VLMs should be trained and evaluated according to document-level correctness rather than only character-level accuracy.

6.5 LightOn OCR

LightOnOCR-1B-1025 — Hugging Face

Architecture

LightOnOCR combines:

- Native-resolution Vision Transformer
- Pixtral/Mistral vision architecture
- Qwen3 language model
- Multimodal projection layer

The projection layer downsamples vision tokens before passing them to the language model.

End-to-End Design

The architecture intentionally avoids complicated multi-stage OCR pipelines.

The basic structure is:

```
Document Image
      ↓
Native Resolution ViT
      ↓
Vision Token Downsampling
      ↓
Qwen3 Decoder
      ↓
Markdown
```

The model does not depend on elaborate OCR preprocessing or multiple specialized models.

Key Innovation

The major research focus is **simplicity and fine-tunability**.

LightOnOCR demonstrates that a compact VLM can be adapted to specialized document domains without requiring a large collection of independent OCR components.

Research Insight

This is particularly important for organizations whose document formats may change over time.

Instead of building:

```
Detector
+ OCR
+ Layout Model
+ Table Model
+ Postprocessor
+ Rules
```

a specialized VLM can potentially be fine-tuned directly on the organization's documents.

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7. Architectural Comparison

Model	Core Architecture	Main Architectural Idea	Primary Strength
PaddleOCR-VL	NaViT + ERNIE	Layout + VLM hybrid	Compact multilingual OCR
Nanonets OCR 2	Qwen2.5-VL	Semantic tagging	Structured document representation
DeepSeek-OCR	SAM + compressor + CLIP + MoE	Visual-token compression	Token efficiency
OlmOCR-2	Qwen2.5-VL	SFT + GRPO	Document-level correctness
LightOn OCR	Pixtral ViT + Qwen3	Simple end-to-end VLM	Fine-tuning and adaptation

8. End-to-End vs Hybrid OCR-VLM

The five systems reveal two major architectural philosophies.

8.1 Mostly End-to-End VLM

Examples:

- Nanonets OCR 2
- OlmOCR-2
- LightOn OCR

General architecture:

```
Full Document Page
        ↓
      VLM
        ↓
Structured Document
```

Advantages

- Preserves global page context
- Uses fewer independent models
- Provides a simpler conceptual architecture
- Supports stronger semantic understanding

- Can directly generate structured output

Challenges

- High vision-token consumption
- GPU memory requirements
- Difficult optimization
- Potential hallucination
- Prompt sensitivity
- Possible inconsistency across document types

8.2 Hybrid VLM Architecture

PaddleOCR-VL provides an example of a hybrid approach:

```
Document
   ↓
Layout Detector
   ↓
Regions
   ├── Text → VLM
   ├── Table → VLM
   ├── Formula → VLM
   └── Chart → VLM
```

This approach separates **where something is** from **what it means**.

Advantages

- More explicit layout control
- Can route different elements to appropriate recognition tasks
- Potentially reduces unnecessary visual processing
- Supports compact VLM deployment

Disadvantages

- Multiple processing stages
- Additional preprocessing
- More pipeline complexity
- Possible context loss between cropped regions

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9. Important Research Findings

Finding 1: OCR is becoming document understanding

Modern OCR-VLMs are no longer limited to:

```
Image → Text
```

The emerging paradigm is:

```
Image
 ↓
Text + Layout + Semantics + Relationships
 ↓
Structured Document
```

This is a fundamental shift from OCR toward **Document Intelligence**.

Finding 2: Layout is as important as text

For resumes, invoices, contracts, research papers, forms, and financial documents, the position of information can change its meaning.

For example:

```
Candidate Name
Job Title
Company
        ↓
Employment Relationship
```

A simple text extraction system may recover all four strings but lose their relationships.

OCR-VLMs attempt to preserve those relationships through:

- Layout understanding
- Reading order
- Grounding
- Semantic tags
- Structured output

Finding 3: Full-page context is valuable

Full-page OCR-VLM approaches demonstrate the importance of maintaining relationships between different document elements.

For example:

```
Table
   ↓
Caption
   ↓
Paragraph explaining table
   ↓
Footnote
```

Processing each component independently can remove this relationship.

A full-page VLM can potentially reason about these elements together.

Finding 4: Token compression is a major research direction

DeepSeek-OCR demonstrates that visual-token compression can become a central component of OCR architecture.

The key research question becomes:

> How much can a document image be compressed before critical document information is lost?

This is especially important for:

- High-resolution PDFs
- Long documents
- Large-scale processing
- GPU-constrained environments

Finding 5: Structured output is more valuable than plain text

Different models demonstrate different structured representations:

```
Markdown
HTML
JSON
LaTeX
Semantic Tags
Bounding Boxes
YAML
```

For downstream document intelligence, structured output can be significantly more useful than raw OCR text.

Finding 6: Training methodology matters

OlmOCR-2 demonstrates that OCR models can benefit from reinforcement learning and document-level verification.

LightOn OCR demonstrates the importance of fine-tuning and compact model design.

Nanonets OCR 2 demonstrates the value of document-specific training and semantic output formatting.

Therefore:

```
Architecture
+
Training Data
+
Training Objective
+
Evaluation Method
```

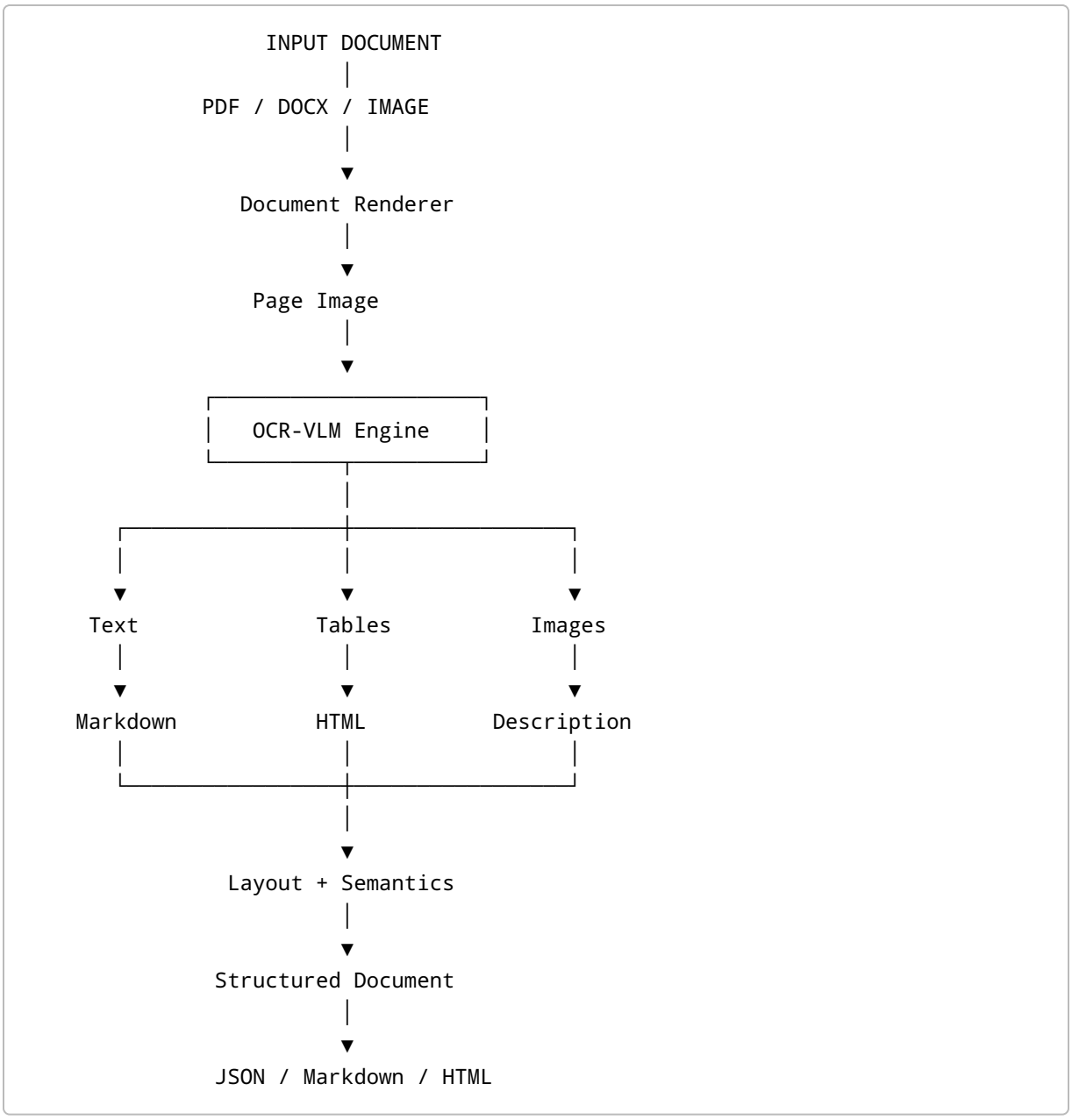
must all be considered together.

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10. Recommended OCR-VLM Architecture for Document Intelligence

Based on the research, a practical architecture should combine the strongest concepts demonstrated by the reviewed systems.

A proposed architecture is:



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11. Recommended Design Principles

Based on this research, the following principles are recommended for an OCR-VLM document intelligence system.

Principle 1 — Preserve full-page context

Whenever possible, process the complete page rather than immediately splitting it into isolated text regions.

Principle 2 — Preserve document structure

The output should retain:

- Headers
- Paragraphs
- Tables
- Lists
- Images
- Captions
- Form fields
- Equations
- Footnotes
- Page numbers

Principle 3 — Use semantic representation

Instead of producing:

```
John Doe
Software Engineer
ABC Company
```

the OCR layer should ideally preserve semantic relationships:

```
{
  "name": "John Doe",
  "job_title": "Software Engineer",
  "company": "ABC Company"
}
```

or an equivalent structured intermediate representation.

Principle 4 — Treat tables as structured objects

Tables should not be flattened into ordinary text.

HTML, Markdown tables, or structured JSON should be considered.

Principle 5 — Preserve mathematical notation

Equations should be represented in LaTeX or another machine-readable mathematical representation.

Principle 6 — Preserve spatial information when required

Bounding boxes and element coordinates are valuable for:

- Forms
- Resumes
- Invoices
- Tables
- Signatures
- Visual verification

Principle 7 — Optimize visual-token usage

High resolution should be used when document detail requires it, but token compression or dynamic resolution should be considered for scalability.

Principle 8 — Evaluate document correctness

Character-level OCR accuracy is insufficient.

Evaluation should include:

- Reading order
- Table structure
- Formula correctness
- Layout preservation
- Semantic correctness
- Image relationships
- Missing-content detection
- Hallucination rate
- Structured-output validity

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12. Proposed Research Direction

The research indicates that a strong OCR-VLM system should not simply select the model with the largest parameter count.

Instead, model selection should be based on the document workload.

For multilingual documents

Consider:

PaddleOCR-VL

because its architecture is designed for broad multilingual document parsing.

For structured business documents

Consider:

Nanonets OCR 2

because semantic tagging is useful for forms, signatures, checkboxes, and downstream LLM processing.

For high-throughput processing

Consider architectural ideas from:

DeepSeek-OCR

because visual-token compression is directly targeted at computational efficiency.

For document-level correctness

Consider:

OlmOCR-2

because its training approach uses programmatic verification and reinforcement learning for difficult OCR cases.

For domain-specific adaptation

Consider:

LightOn OCR

because its compact end-to-end architecture is designed to be comparatively straightforward to fine-tune.

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13. Recommended Experimental Evaluation

Before selecting one model for production, the models should be tested against a common internal dataset.

The evaluation dataset should contain:

```
1. Normal digital PDFs
2. Scanned PDFs
3. Multi-column documents
4. Tables
5. Complex tables
6. Forms
7. Handwritten text
8. Mathematical equations
9. Images with captions
10. Charts
11. Multilingual documents
12. Low-quality scans
13. Small-font documents
14. Resumes
15. Legal documents
16. Financial documents
```

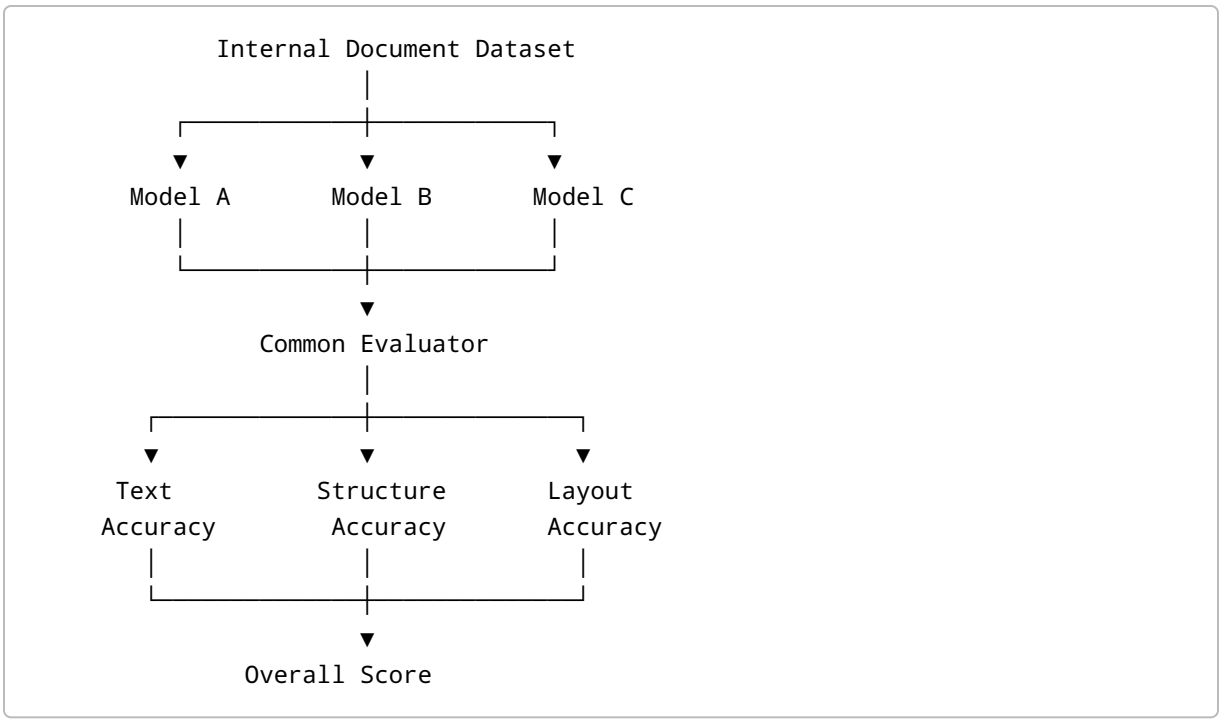
Each model should be evaluated on:

Metric	Purpose
Text accuracy	Basic OCR quality
Reading-order accuracy	Correct document sequence
Table accuracy	Structural table extraction
Formula accuracy	Mathematical representation
Layout accuracy	Spatial structure
Semantic accuracy	Correct element interpretation
Missing-content rate	Completeness
Hallucination rate	Prevent invented content
Structured-output validity	Machine-processable output
Inference efficiency	Operational feasibility

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14. Proposed Benchmark Strategy

Rather than selecting a model based solely on published benchmark numbers, the recommended process is:



This will provide a more reliable basis for selecting a production OCR-VLM.

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15. Conclusion

The research shows that modern OCR is evolving from a collection of specialized recognition components into **multimodal document understanding systems**.

The five models examined demonstrate different solutions to the same fundamental problem:

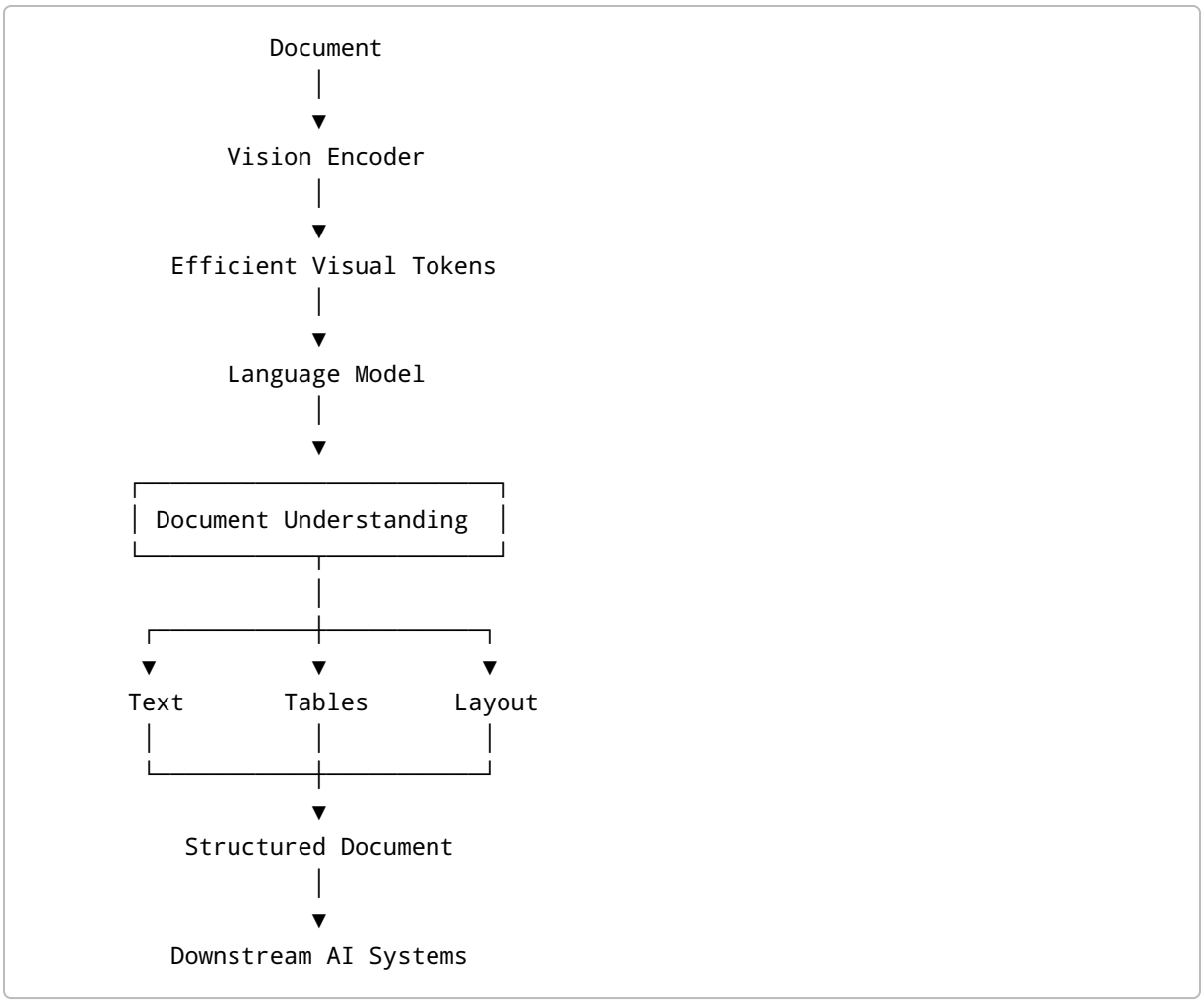
> How can a machine convert a visually complex document into an accurate, structured, machine-readable representation while preserving both content and context?

The major research directions are:

1. **Compact multilingual VLMs** — demonstrated by PaddleOCR-VL.
2. **Semantic document tagging** — demonstrated by Nanonets OCR 2.
3. **Vision-token compression** — demonstrated by DeepSeek-OCR.
4. **Document-level reinforcement learning** — demonstrated by OlmOCR-2.
5. **Simple domain-adaptable VLMs** — demonstrated by LightOn OCR.

The strongest conclusion is that **OCR-VLM should be treated as a document understanding layer rather than merely an OCR engine**.

For a production document intelligence platform, the target output should therefore be:



This architecture provides a strong foundation for applications such as **resume parsing, document search, document question answering, contract analysis, invoice processing, financial-document extraction, and enterprise document intelligence**.

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16. Research Sources

Model Documentation

- PaddleOCR-VL — Hugging Face
- Nanonets OCR 2 — Hugging Face
- DeepSeek-OCR — Hugging Face
- OlmOCR-2 — Hugging Face
- LightOnOCR-1B-1025 — Hugging Face

The model-specific architecture claims in this report should be validated against the latest official model documentation before production adoption.

Final Research Position

OCR-VLM is best understood as the convergence of OCR, layout analysis, visual reasoning, and structured document generation.

The most promising research direction is therefore not simply:

"Which OCR model has the highest OCR score?"

but:

"Which architecture provides the best balance of recognition accuracy, document structure preservation, contextual understanding, visual-token efficiency, and adaptability for our specific document workload?"